



Designing a DEER-Toolkit Enabling Educational Escape Rooms in a 2D Dungeon Crawler Game

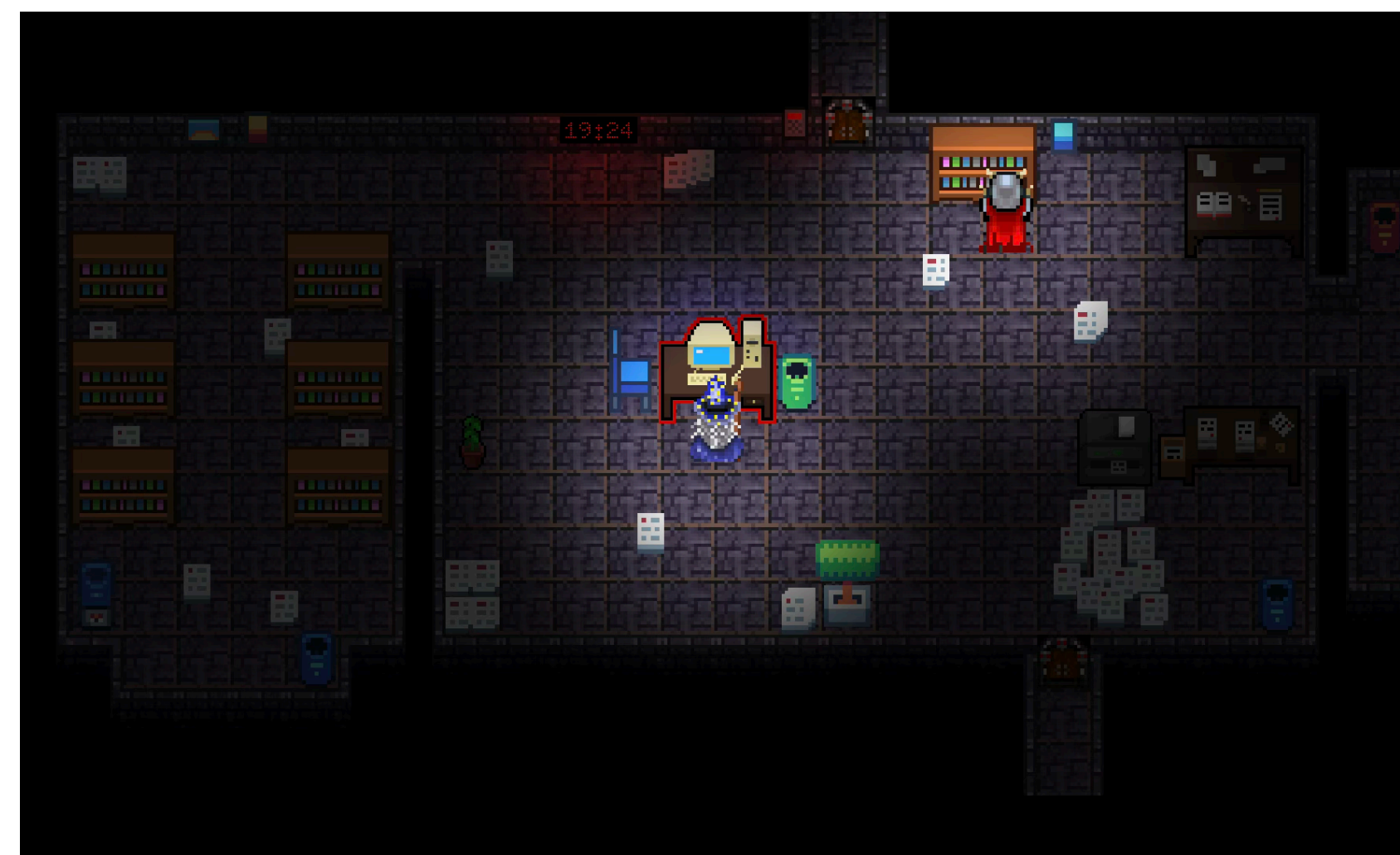


Figure 1: The CyberRoom level. Players start in an office without electricity.



Figure 2: Carelessly discarded login credentials in the office bin.

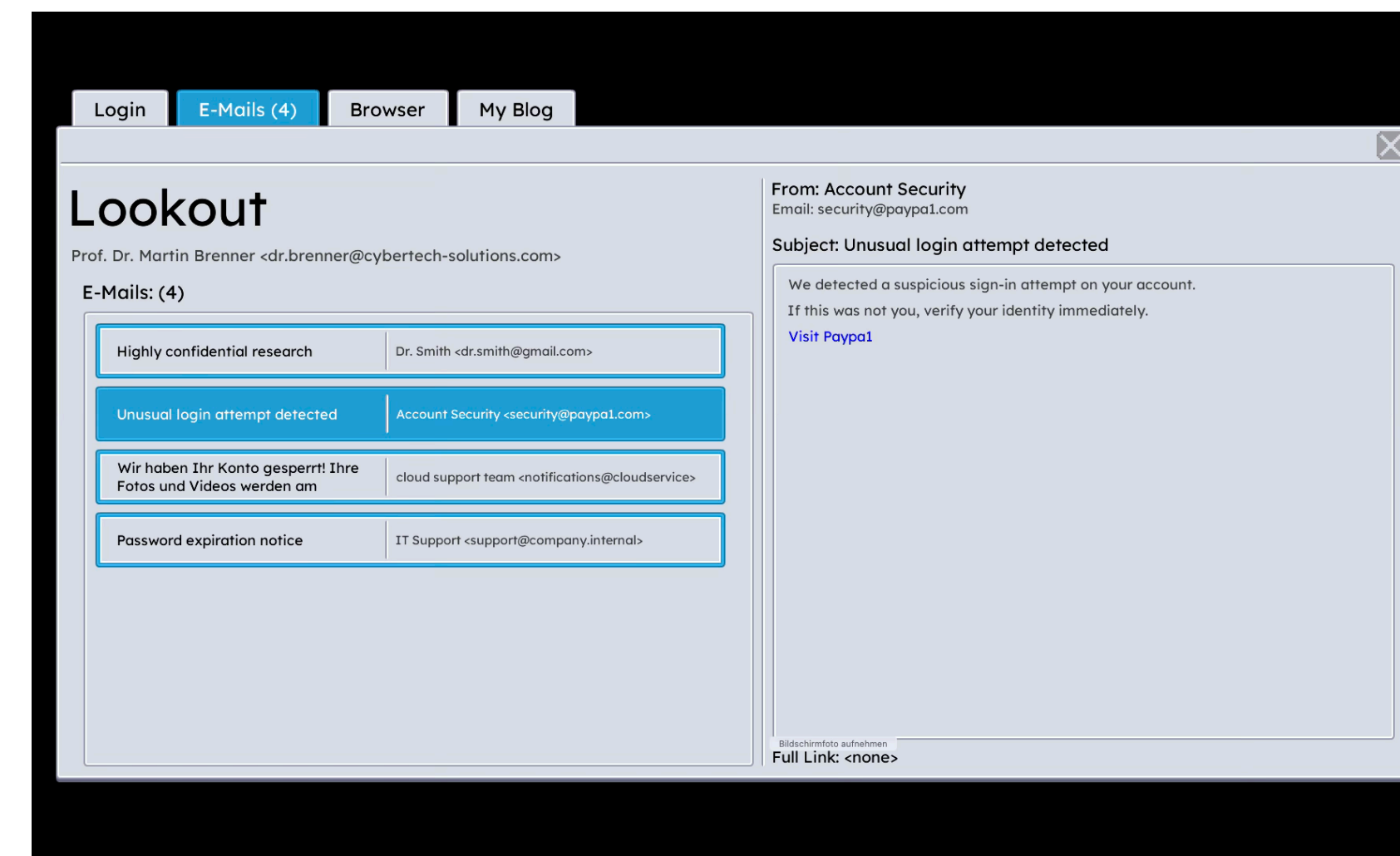


Figure 3: The email inbox - players must identify the relevant message among spam.

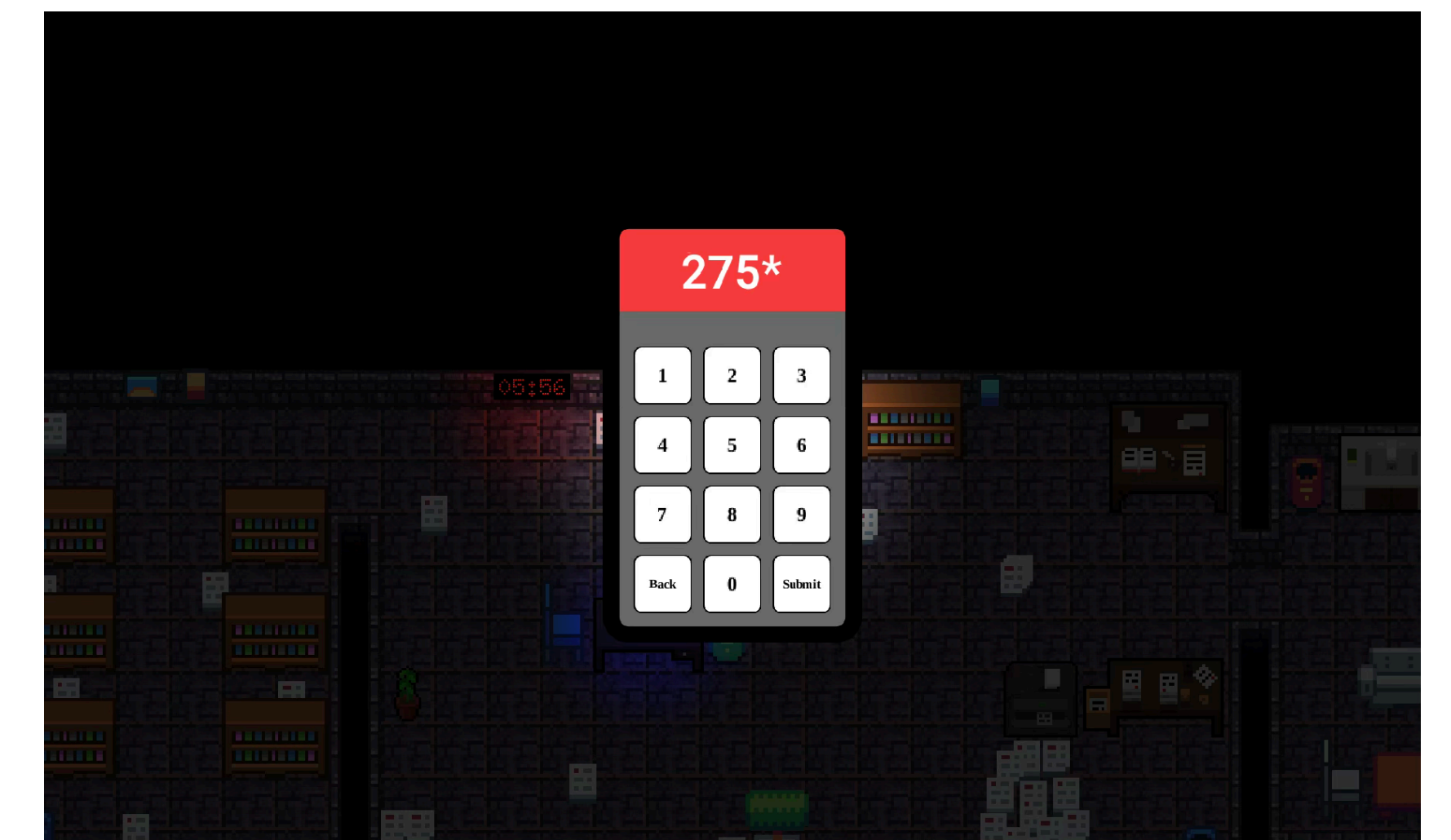


Figure 4: The door keypad, unlocked by decoding a Morse code message.

Demonstrator: "The Last Hour"

A scientist has gone missing. Players must investigate and escape before time runs out. This 10-minute cybersecurity Digital Educational Escape Room (DEER) for 1-2 players is built around a chain of interconnected puzzles:

1. Players enter a **dark office** with no power (Fig. 1). They search the environment, locate the main switch, and restore electricity.
2. The PC is **locked**. Players find carelessly discarded login credentials in the bin (Fig. 2), learning first-hand about secure password disposal.
3. Logged in, they face an **inbox full of emails** including spam (Fig. 3). An internet blog provides clues to identify the relevant message.
4. The correct email links to a website containing an **encrypted code**. Cross-referencing the blog entry reveals which code must be deciphered.
5. Decoding the **ASCII cipher** yields a document containing a message in Morse code.
6. The decoded Morse reveals the combination for the **door keypad** (Fig. 4). Players escape, and the debrief connects every puzzle step to real cybersecurity practices.

DEER-Toolkit

DEERs are a form of game-based learning in which teams collaboratively solve interconnected puzzles to achieve shared learning objectives [3]. Empirical research shows that DEERs **increase learner motivation**, foster collaboration, and promote deeper cognitive engagement compared to conventional formats [2, 4]. Building on the open-source Dungeon framework developed at Hochschule Bielefeld (Germany), we created the **DEER-Toolkit**: a toolbox embedded in the Dungeon that provides reusable building blocks for DEER scenarios. It combines accessibility, interactivity, and educational value, supporting lightweight, low-hardware setups. Core toolkit components are:

- **Puzzle mechanics**: Standardized interaction primitives (locks, switches, discovery tasks) supporting logic, coordination, search, and communication challenges [1].
- **Adaptive hints**: A progressive hint system.
- **Multiplayer**: Cooperative play [2].
- **Learning analytics**: Actions are logged via xAPI to a Learning Record Store. Real-time instructor dashboards support facilitation and targeted debriefing [5].
- **Immersion**: Thematic soundscapes and contextual cues enhance atmosphere and environmental storytelling.

Future Work

Our roadmap focuses on three directions:

1. **Expand the toolkit** by adding puzzle types, mechanics, and narrative templates to broaden expressive coverage and ease of use.
2. **Implement a low-code editor**: Creating DEERs still requires substantial technical expertise, which limits adoption among educators. We are therefore developing a **visual low-code editor**. Authors design rooms, puzzles, and story beats through drag-and-drop tools without any scripting. The key question is whether non-technical educators can design coherent DEERs without external support.
3. **Real-world partnerships**: We are in active discussions with nursing care professionals to co-design domain-specific escape rooms. We are open for more collaborations.

OER and OSS principles remain foundational throughout, ensuring all tools, scenarios, and findings are openly accessible and adaptable.

Literature
[1] Nicholson, S. (2016). The State of Escape: Escape Room Design and Facilities. Lansing.
[2] Veldkamp, A. et al. (2020). Escape education: A systematic review. "Educational Research Review, 31."
[3] Nicholson, S. (2015). Peeking behind the locked door: A survey of escape room facilities.
[4] Veldkamp, A. et al. (2022). You Escaped! How Did You Learn during Gameplay? "British Journal of Educational Technology, 53"(5).
[5] Queiroz, R. et al. (2023). Integrating Gamified Educational Escape Rooms in Learning Management Systems. "OASICS, SLATE 2023."

